

Parent-Co-Teacher Demo Walkthrough

Gemini 3 API Hackathon Submission

This document captures the complete demo flow with screenshots and explanations for each step.

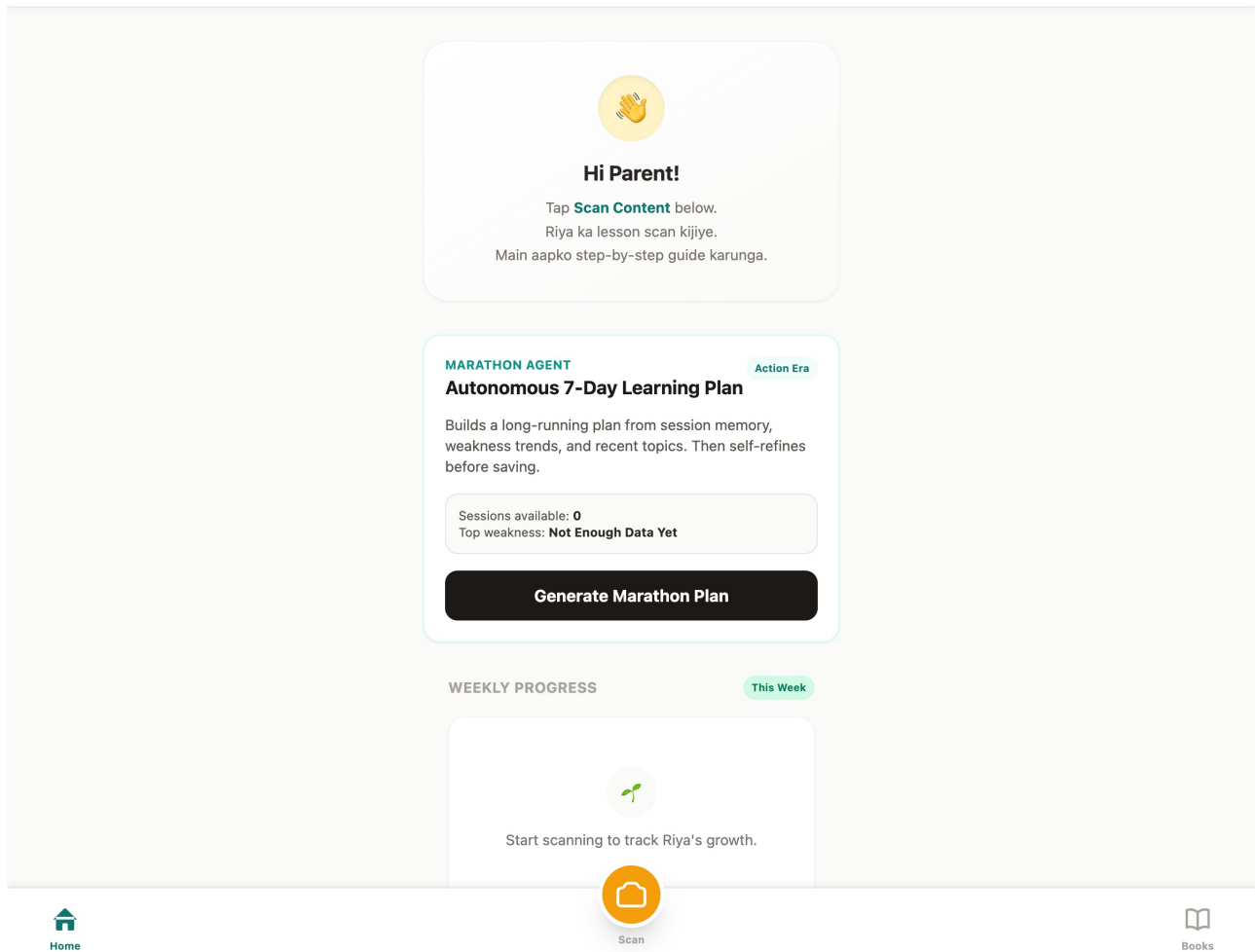
Overview

Parent-Co-Teacher is an AI-powered learning orchestration system that helps parents with limited English literacy teach their children effectively using Hinglish (natural Hindi-English code-mixing) and audio guidance.

Key Gemini Integrations:

- Gemini 3 Pro (Vision) - Homework image analysis
 - Gemini 3 Pro (Reasoning) - Marathon Agent with self-critique
 - Gemini 3 Flash - Bilingual content generation
 - Gemini TTS - Audio synthesis for accessibility
-

Step 1: Home Screen (Fresh State)



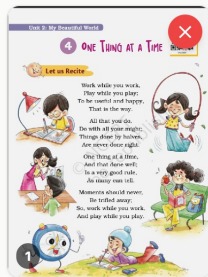
What you see:

- Clean dashboard with "Hi Parent!" greeting in bilingual text
- Kid profile selector (Riya - Class 4, English)
- Marathon Agent panel showing "Sessions available: 0"
- Scan button prominently displayed at bottom

Key talking point: "The app greets parents in both English and Hinglish from the start, making them feel comfortable."

Step 2: Scan Homework

Scanned Pages



1 page scanned

Start Teaching

+ Add Another Page

What you see:

- Uploaded homework image preview (English poem: "One Thing at a Time")
- Page counter showing "1 page scanned"
- "Start Teaching" button to begin analysis
- Option to add more pages

Key talking point: "Parents simply photograph their child's homework. No typing required."

Step 3: Chapter Guide Generated

One Thing at a Time

PART 1/4

×

BOOK TEXT

"Work while you work, Play while you play;
To be useful and happy, That is the way."

▶ Read Aloud

🧠 Understand It

Is stanza ka simple matlab hai ki jab kaam ka waqt ho toh sirf kaam karo, aur jab khelne ka waqt ho toh sirf khelo. Tabhi hum khush reh sakte hain.

▶ Listen (Hinglish)

🗣️ Say to Riya

"Riya, iska matlab hai jab aap padhai kar rahe ho, toh bas padhai karo. Aur jab play time ho, toh full enjoy karo. Dono ko mix mat karo, tabhi aap happy rahoge."

▶ Play Script

Next Part →

What you see:

- **Book Text** (cream card): Original poem extracted by Gemini Vision
- **Understand It** (yellow card): Hinglish explanation for the parent
 - "Is stanza ka simple matlab hai ki jab kaam ka waqt ho toh sirf kaam karo..."
- **Say to Riya** (teal card): Exact script parent speaks to child
 - "Riya, iska matlab hai jab aap padhai kar rahe ho, toh bas padhai karo..."
- Audio buttons for each section

Key talking point: "This isn't translation - it's natural Hinglish, the way Indians actually speak. The parent knows exactly what to say."

Step 4: Audio Playback

One Thing at a Time

PART 1/4

BOOK TEXT

*"Work while you work, Play while you play;
To be useful and happy, That is the way."*

▶ Read Aloud

🧠 Understand It

Is stanza ka simple matlab hai ki jab kaam ka waqt ho toh sirf kaam karo, aur jab khelne ka waqt ho toh sirf khelo. Tabhi hum khush reh sakte hain.

▶ Listen (Hinglish)

🗣️ Say to Riya

"Riya, iska matlab hai jab aap padhai kar rahe ho, toh bas padhai karo. Aur jab play time ho, toh full enjoy karo. Dono ko mix mat karo, tabhi aap happy rahoge."

▶ Play Script

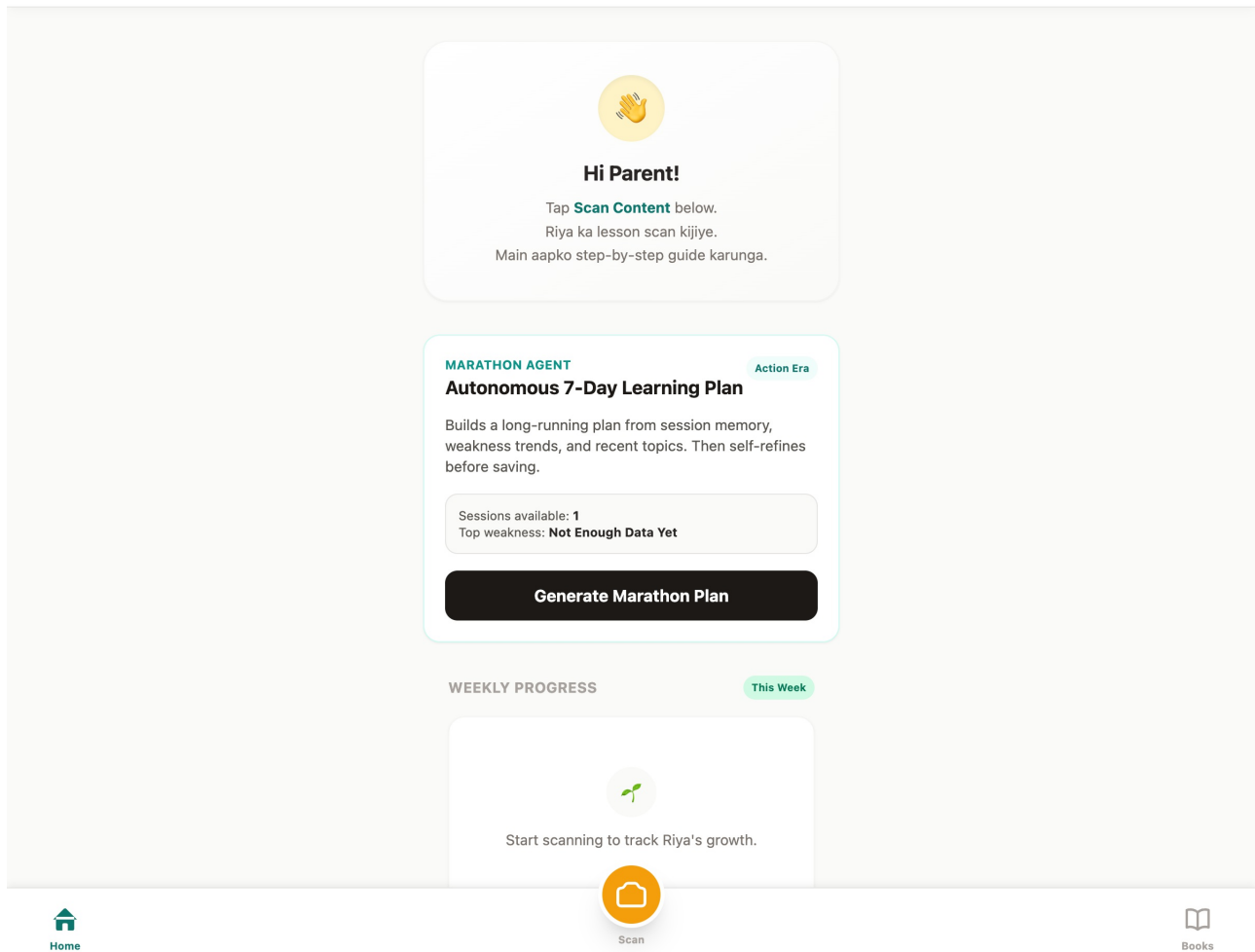
Next Part →

What happens:

- Clicking "Play Script" generates TTS audio via Gemini
- Parent hears the Hinglish script spoken naturally
- Button changes to "Stop" during playback
- Visual feedback confirms audio is playing

Key talking point: "Even if the parent struggles to read, they can LISTEN to the script and repeat it to their child. This is accessibility in action."

Step 5: Home After Session

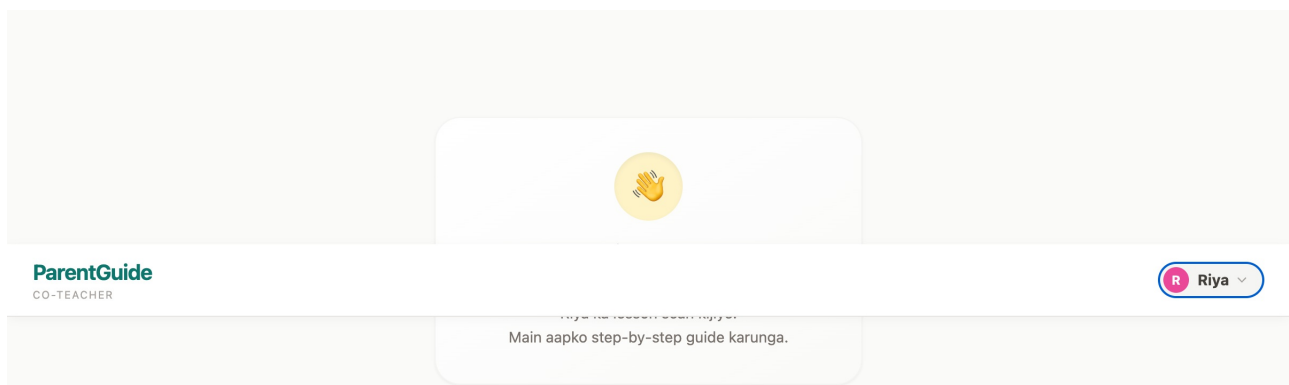


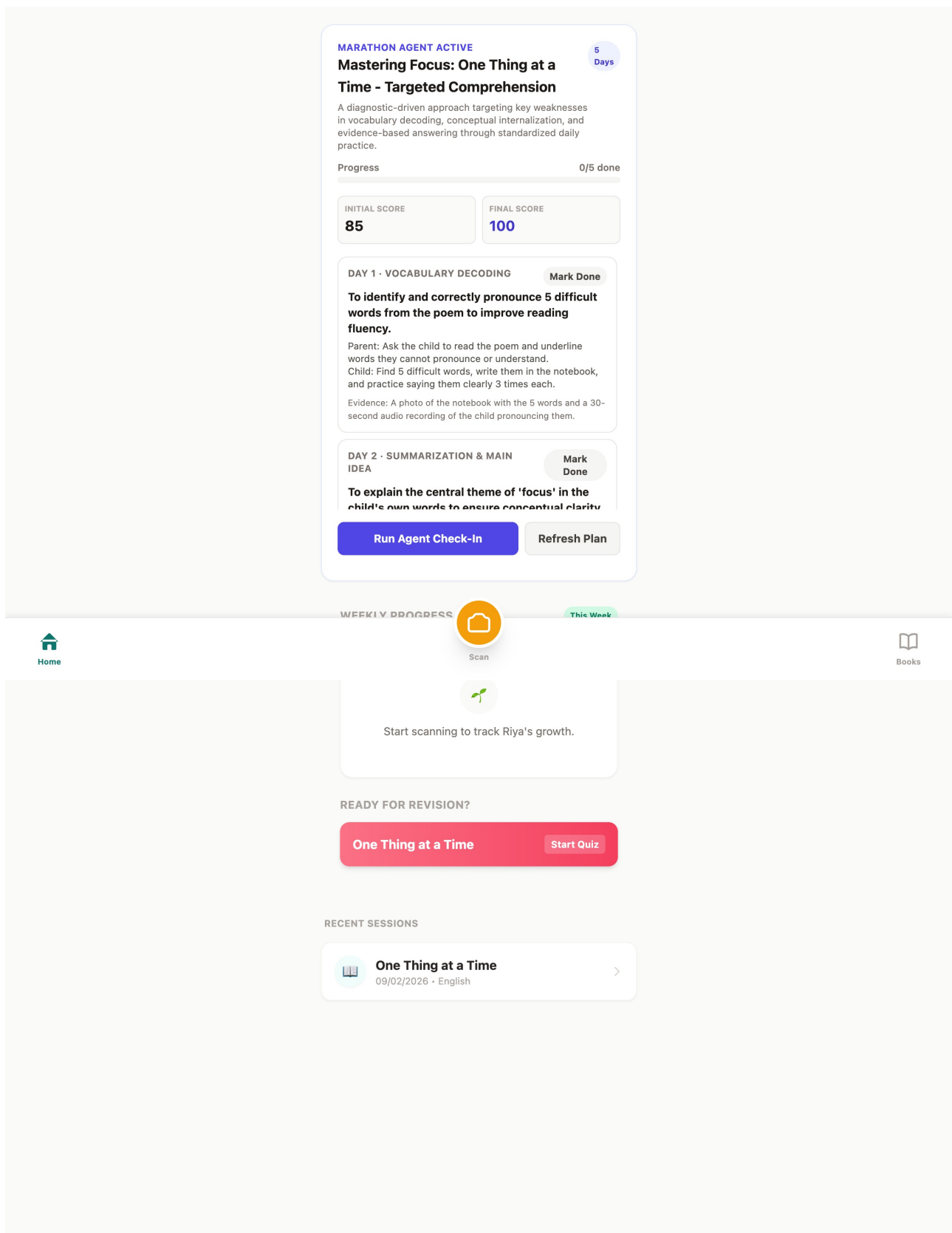
What you see:

- "Sessions available: 1" - Session was tracked
- "Ready for Revision?" section appeared with quiz option
- "Recent Sessions" shows saved session
- All data persisted in IndexedDB

Key talking point: "The app remembers every session. This isn't a stateless chatbot - it's building a learning history."

Step 6: Marathon Agent Generated





What you see:

- "Marathon Agent Active" header with 5 Days badge
- **Quality Scores: Initial 85 → Final 100** (self-improvement!)
- **5-Day Learning Plan** with structured missions:
 - Day 1: Vocabulary Decoding

- Day 2: Summarization & Main Idea
 - Day 3: Visual Processing
 - Day 4: Textual Analysis
 - Day 5: Critical Thinking & Response
- Each mission has:
 - Parent action
 - Child task
 - Evidence requirement (photo proof)

Key talking point: "Notice the quality score improvement - the agent critiqued its own plan and improved it from 85 to 100. This is autonomous reasoning, not a simple prompt."

Step 7: Adaptive Check-In

Hi Parent!

Tap **Scan Content** below.
 Riya ka lesson scan kijiye.
 Main aapko step-by-step guide karunga.

MARATHON AGENT ACTIVE 5 Days

Mastering Focus: One Thing at a Time - Targeted Comprehension

A diagnostic-driven approach targeting key weaknesses in vocabulary decoding, conceptual internalization, and evidence-based answering through standardized daily practice.

Progress 1/5 done

INITIAL SCORE	FINAL SCORE
85	100

ParentGuide
CO-TEACHER

R Riya

DAY 5 · CRITICAL THINKING & RESPONSE Mark Done

To answer inferential questions accurately to prepare for examinations.

Parent: Ask two specific 'Why' questions: 'Why should we not play while we work?' and 'How does doing one thing at a time help us?'

Child: Write the answers to these two questions in complete sentences using evidence from the poem.

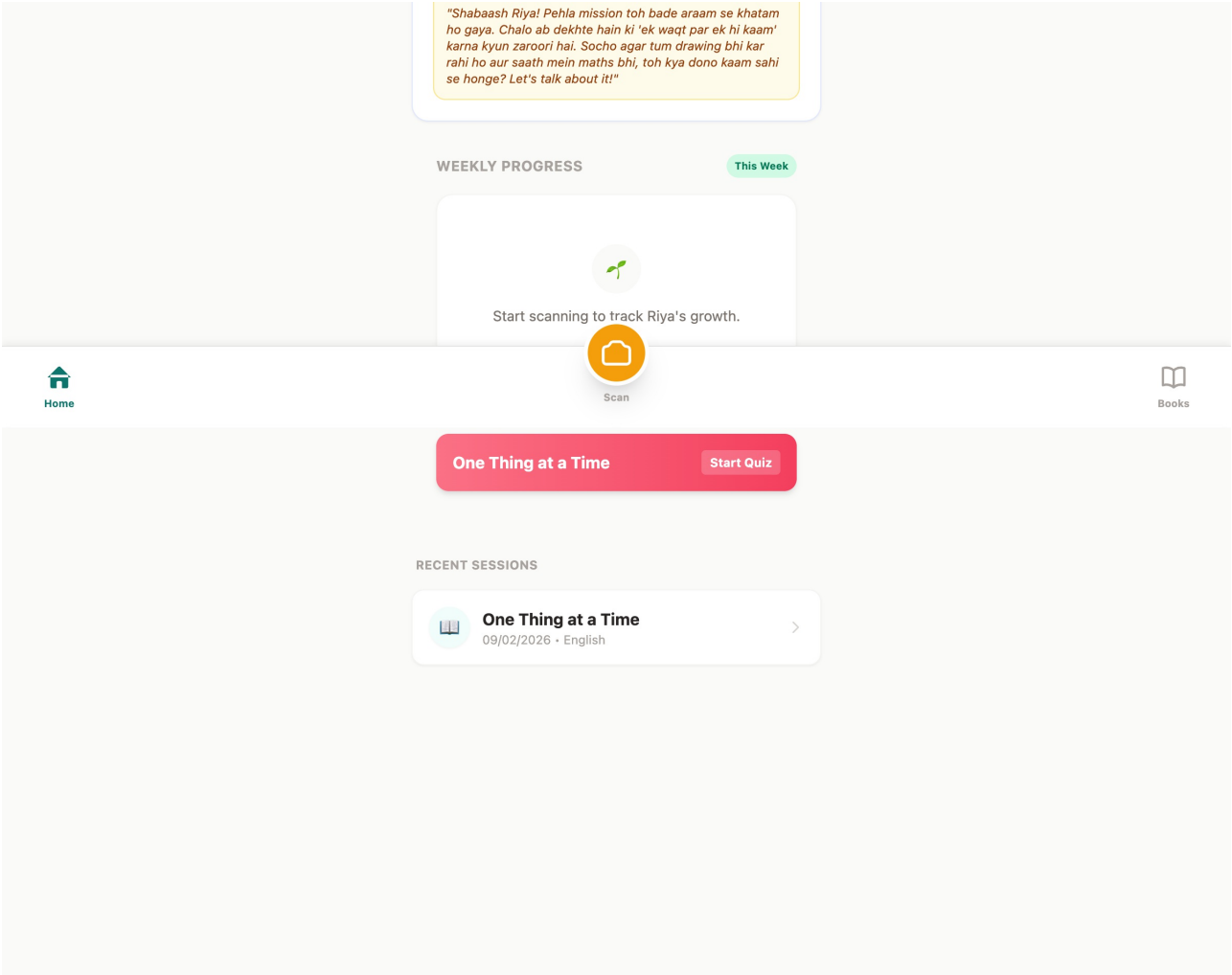
Evidence: A photo of the written answers to the two inferential questions.

Run Agent Check-In Refresh Plan

LATEST AGENT CHECK-IN

Riya has successfully completed the first mission. The focus now shifts to conceptualizing the theme of the poem and understanding the importance of concentration through discussion and creative expression.

Start mission m-2 to explore the concept of 'focus' and the poem's central message.



What you see:

- Progress updated to "1/5 done" after marking Day 1 complete
- **Latest Agent Check-In** section (cream box) showing:
 - Summary: "Riya has successfully completed the first mission..."
 - Next action: "Start mission m-2 to explore the concept of 'focus'..."
 - **Hinglish motivation script:** "Shabaash Riya! Pehla mission toh bade araam se khatam ho gaya. Chalo ab dekhte hain ki 'ek waqt par ek hi kaam' karna kyun zaroori hai..."
- Adjusted missions based on completed work

Key talking point: "The agent learns from progress and adapts. It motivates in Hinglish and adjusts the remaining plan. This is an orchestrated system that evolves."

Technical Highlights for Judges

Gemini Integration Depth (Technical Execution - 40%)

Feature	Gemini API Used	Description
Image Analysis	Gemini 3 Pro Vision	Detects subject, chapter, content type
Parent Guide	Gemini 3 Pro	Generates bilingual teaching scripts
Marathon Planning	Gemini 3 Pro (2-pass) Draft → Self-critique → Refine	

Adaptive Check-ins Gemini 3 Pro

Analyzes progress, adjusts missions

Audio TTS

Gemini TTS

Converts scripts to natural speech

Why This Isn't a Prompt Wrapper (Innovation - 30%)

- **Stateful:** IndexedDB persistence across sessions
- **Multi-step pipeline:** Vision → Analysis → Generation → TTS
- **Self-improving:** Quality scoring with verification checklist
- **Adaptive:** Check-ins modify future plans based on progress
- **Orchestrated:** 6+ Gemini API calls per session, coordinated

Real-World Impact (Impact - 20%)

- **250 million+ Indian parents** face language barriers with English-medium schools
 - **Audio-first design** enables parents regardless of reading ability
 - **Hinglish code-mixing** matches natural Indian communication patterns
 - **Evidence-based tracking** creates accountability and measurable progress
-

Demo Tips

1. **Lead with emotion:** "Meet Sunita, a house helper in Mumbai..."
 2. **Play the audio:** The Hinglish TTS is your strongest differentiator
 3. **Show quality scores:** 85 → 100 is visual proof of autonomous reasoning
 4. **Don't apologize:** Confidence matters
-

Quick Commands for Fresh Demo

```
# To clear data and start fresh:  
# Open DevTools → Application → IndexedDB → Delete "ParentTeacherDB"  
  
# Or refresh and the app will be ready
```

Good luck with your submission!

Built with Gemini 3 API for the Gemini 3 Hackathon 2025